Chemical Bonding and Molecular Structure

Syllabus Context: Physical & Inorganic Chemistry Foundation — Kössel-Lewis Approach, Octet Rule, Ionic Bonding & Lattice Enthalpy, Bond Parameters, Resonance, Dipole Moment, Fajans' Rules, VSEPR Theory, Valence Bond Theory, Hybridisation, Molecular Orbital Theory (LCAO), and Hydrogen Bonding.

1. Learning Objectives & Chapter Overview

After studying this tutorial, you will be able to:

- Understand the **Kössel-Lewis approach** to chemical bonding, Lewis symbols, and group valence.
- Explain the **octet rule** and its limitations (incomplete octet, odd-electron molecules, and expanded octet), and draw Lewis structures and formal charges of simple molecules and polyatomic ions.
- Describe the formation of **electrovalent (ionic) bonds**, favourable energetic factors, and the role of **lattice enthalpy** in crystal stability.
- Define and correlate **bond parameters** (bond length, covalent vs. van der Waals radius, bond angle, bond dissociation enthalpy, average bond enthalpy, bond order, resonance, and bond polarity).
- Predict the geometrical shapes of covalent molecules and ions using the **Valence Shell Electron Pair Repulsion (VSEPR) theory**.
- Explain the **Valence Bond (VB) theory**, orbital overlap criteria, directional properties of bonds, and distinguish between sigma (σ) and pi (π) bonds.
- Analyse different types of **hybridisation** involving s, p, and d orbitals (sp, sp^2, sp^3, dsp^2, sp^3d, sp^3d^2, d^2sp^3) and deduce molecular geometries.
- Apply the **Molecular Orbital (MO) theory** to homonuclear diatomic molecules to determine electronic configurations, stability, bond order, bond length, and magnetic behaviour.
- Explain the cause and types of **hydrogen bonding** (intermolecular and intramolecular) and its influence on physical properties.

Overview: Under normal conditions, no element exists as an independent atom in nature except noble gases. Instead, groups of atoms exist together as one species having characteristic properties, called a *molecule*. **The attractive force which holds various constituents (atoms, ions, etc.) together in different chemical species is called a chemical bond.** Every system in nature tends to become more stable, and chemical bonding is nature's way of lowering the potential energy of a system to attain stability.

2. Kössel-Lewis Approach to Chemical Bonding

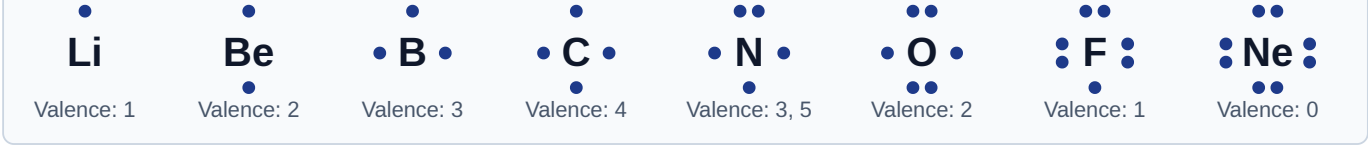
In 1916, W. Kössel and G.N. Lewis independently provided the first logical explanation of valence based on the chemical inertness of noble gases.

- **Lewis's Picture of the Atom:** Lewis pictured the atom in terms of a positively charged **'Kernel'** (the nucleus plus the inner shell electrons) and an outer shell that could accommodate a maximum of eight electrons occupying the eight corners of a cube surrounding the kernel. An octet of outer electrons represents a particularly stable electronic arrangement.
- **Lewis's Postulate:** Atoms achieve a stable octet when they are linked by chemical bonds—either by transfer of electrons (as in Na^+ and Cl^-) or by sharing a pair of electrons between atoms (as in $\mathrm{Cl}_2, \mathrm{H}_2, \mathrm{F}_2$).

2.1 Lewis Symbols and Group Valence

Only outer-shell electrons (**valence electrons**) take part in chemical combination; inner-shell electrons are well protected. **Lewis symbols** are simple notations representing valence electrons as dots around the chemical symbol of an element.

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Tutorial Diagram 1: Lewis dot symbols and common group valences for second-period elements.

Significance of Lewis Symbols: The number of dots around the symbol represents the number of valence electrons. The **common or group valence** of an element is generally equal either to the *number of dots* in its Lewis symbol or to $8 - \text{number of dots}$.

2.2 Kössel's Postulations & Electrovalent Bond

Kössel highlighted the following key facts regarding chemical bonding:

1. In the periodic table, highly electronegative halogens and highly electropositive alkali metals are separated by noble gases.
2. Formation of a negative ion from a halogen atom and a positive ion from an alkali metal atom is associated with the gain and loss of an electron by the respective atoms.
3. The negative and positive ions thus formed attain stable noble gas electronic configurations (ns^2np^6, an octet of electrons, except helium which has a duplet $1s^2$).
4. The negative and positive ions are stabilized by electrostatic attraction.

Formation of NaCl:

$$\mathrm{Na}\left([\mathrm{Ne}]3s^1\right) \rightarrow \mathrm{Na}^+\left([\mathrm{Ne}]\right) + \mathrm{e}^-$$
$$\mathrm{Cl}\left([\mathrm{Ne}]3s^2 3p^5\right) + \mathrm{e}^- \rightarrow \mathrm{Cl}^-\left([\mathrm{Ne}]3s^2 3p^6 \text{ or } [\mathrm{Ar}]\right)$$
$$\mathrm{Na}^+ + \mathrm{Cl}^- \rightarrow \mathrm{NaCl} \text{ or } \mathrm{Na}^+\mathrm{Cl}^-$$

Formation of $\mathrm{CaF_2}$:

$$\mathrm{Ca}\left([\mathrm{Ar}]4s^2\right) \rightarrow \mathrm{Ca}^{2+}\left([\mathrm{Ar}]\right) + 2\mathrm{e}^-$$
$$\mathrm{F}\left([\mathrm{He}]2s^2 2p^5\right) + \mathrm{e}^- \rightarrow \mathrm{F}^-\left([\mathrm{He}]2s^2 2p^6 \text{ or } [\mathrm{Ne}]\right)$$
$$\mathrm{Ca}^{2+} + 2\mathrm{F}^- \rightarrow \mathrm{CaF_2} \text{ or } \mathrm{Ca}^{2+}(\mathrm{F}^-)_2$$

The bond formed as a result of electrostatic attraction between positive and negative ions was termed the **electrovalent bond**. The **electrovalence** is equal to the number of unit charge(s) on the ion (e.g., calcium has a positive electrovalence of two, while chlorine has a negative electrovalence of one).

2.3 Octet Rule and Covalent Bond

Octet Rule (Electronic Theory of Chemical Bonding, 1916): Atoms can combine either by transfer of valence electrons from one atom to another (gaining or losing) or by sharing of valence electrons in order to have an octet in their valence shells.

Covalent Bond (Langmuir, 1919): Langmuir refined Lewis's postulations by abandoning the stationary cubical arrangement of the octet and introducing the term *covalent bond*. According to the Lewis-Langmuir theory, a covalent bond is formed when atoms share one or more electron pairs, with each combining atom contributing at least one electron to the shared pair so that both attain outer-shell noble gas configurations.

- **Single Covalent Bond:** Sharing of one electron pair between two atoms (e.g., $\mathrm{Cl_2}$, $\mathrm{H_2O}$, $\mathrm{CCl_4}$).
- **Double Bond:** Sharing of two electron pairs between two atoms (e.g., $\mathrm{CO_2}$ has two $\mathrm{C=O}$ double bonds; ethene $\mathrm{C_2H_4}$ has a $\mathrm{C=C}$ double bond).
- **Triple Bond:** Sharing of three electron pairs between two atoms (e.g., $\mathrm{N_2}$ has $:\mathrm{N \equiv N}:$; ethyne $\mathrm{C_2H_2}$ has $\mathrm{H-C \equiv C-H}$).

2.4 Steps for Writing Lewis Dot Structures

1. **Count total valence electrons:** Add the valence electrons of all combining atoms (e.g., $\mathrm{CH_4}$ has $4 + 4 \times 1 = 8$ valence electrons).
2. **Adjust for ionic charge:** For anions, add one electron per unit negative charge (e.g., add $2\mathrm{e}^-$ for $\mathrm{CO_3^{2-}}$). For cations, subtract one electron per unit positive charge (e.g., subtract $1\mathrm{e}^-$ for $\mathrm{NH_4^+}$).
3. **Set up the skeletal structure:** Place the *least electronegative atom* in the central position (hydrogen and fluorine are always terminal). For example, in $\mathrm{NF_3}$ and $\mathrm{CO_3^{2-}}$, N and C are central atoms.

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4. **Distribute electrons:** Place one shared electron pair for each single bond first. Use remaining electron pairs either for multiple bonding or as lone pairs so that each bonded atom gets an octet of electrons (and H gets a duplet).

Molecule / Ion	Total Valence Electrons	Lewis Representation & Bonding Description
$\mathrm{H_2}$	$1+1=2$	$\mathrm{H:H}$ or $\mathrm{H-H}$ (Each H atom attains helium duplet configuration)
$\mathrm{O_2}$	$6+6=12$	$:\ddot{\mathrm{O}}::\ddot{\mathrm{O}}:$ or $:\ddot{\mathrm{O}}=\ddot{\mathrm{O}}:$ (One double bond; two lone pairs on each O)
$\mathrm{O_3}$	$3\times 6=18$	$:\ddot{\mathrm{O}}=\ddot{\mathrm{O}}^{+}-\ddot{\mathrm{O}}:^{-}$ (Central O has 1 lone pair, double-bonded O has 2 lone pairs, single-bonded O has 3 lone pairs)
$\mathrm{NF_3}$	$5+3(7)=26$	Central N has 1 lone pair and three single bonds to F; each F has 3 lone pairs
$\mathrm{CO_3^{2-}}$	$4+3(6)+2=24$	Central C has no lone pair, one $\mathrm{C=O}$ double bond, and two $\mathrm{C-O^-}$ single bonds
$\mathrm{HNO_3}$	$1+5+3(6)=24$	$\mathrm{H-\ddot{O}-N^{+}(=\ddot{O})-\ddot{O}:^{-}}$ (Central N has no lone pair, bonded to OH, $=\mathrm{O}$, and $-\mathrm{O^-}$)

2.5 Formal Charge

In polyatomic ions, the net charge belongs to the ion as a whole, but we can assign a **formal charge (F.C.)** to each atom. It is defined as the difference between the number of valence electrons of that atom in an isolated/free state and the number of electrons assigned to that atom in the Lewis structure (assuming each atom owns one electron of each shared pair and both electrons of its lone pairs).

$$\text{Formal Charge (F.C.)} = \begin{bmatrix}\text{Total valence } \mathrm{e^-}\\ \text{in free atom}\end{bmatrix} - \begin{bmatrix}\text{Total non-bonding}\\ \text{(lone pair) } \mathrm{e^-}\end{bmatrix} - \frac{1}{2}\begin{bmatrix}\text{Total bonding}\\ \text{(shared) } \mathrm{e^-}\end{bmatrix}$$

Example — Ozone ($\mathrm{O_3}$): In $\ddot{\mathrm{O}}_{(2)}=\ddot{\mathrm{O}}_{(1)}-\ddot{\mathrm{O}}_{(3)}:$

- Central O atom (marked 1): $\mathrm{F.C.}=6-2-\frac{1}{2}(6)=+1$
- Doubly bonded end O atom (marked 2): $\mathrm{F.C.}=6-4-\frac{1}{2}(4)=0$
- Singly bonded end O atom (marked 3): $\mathrm{F.C.}=6-6-\frac{1}{2}(2)=-1$

Note: Formal charges do not indicate real charge separation within the molecule. They help keep track of valence electrons and select the **lowest energy structure** (which is generally the one with the **smallest formal charges** on the atoms).

2.6 Limitations of the Octet Rule

Although useful for second-period elements and organic compounds, the octet rule is not universal:

1. **Incomplete Octet of the Central Atom:** Occurs in compounds where the central atom has fewer than eight electrons around it, especially elements with fewer than four valence electrons ($\mathrm{Li, Be, B}$ having 1, 2, 3 valence electrons). Examples: LiCl (2 electrons), $\mathrm{BeH_2}$ (4 electrons), $\mathrm{BCl_3}$, $\mathrm{BF_3}$, $\mathrm{AlCl_3}$ (6 electrons around central atom).
2. **Odd-Electron Molecules:** In molecules with an odd total number of valence electrons, the octet rule cannot be satisfied for all atoms. Examples: Nitric oxide (NO, 11 valence $\mathrm{e^-}$: $\cdot\ddot{\mathrm{N}}=\ddot{\mathrm{O}}:$) and nitrogen dioxide ($\mathrm{NO_2}$, 17 valence $\mathrm{e^-}$: $:\ddot{\mathrm{O}}=\dot{\mathrm{N}}^{+}-\ddot{\mathrm{O}}:^{-}$).
3. **The Expanded Octet:** Elements in and beyond the third period of the periodic table have $3d$ orbitals available in addition to $3s$ and $3p$ orbitals for bonding, allowing more than eight valence electrons around the central atom. Examples: $\mathrm{PF_5}$ (10 electrons around P), $\mathrm{SF_6}$ (12 electrons around S), $\mathrm{H_2SO_4}$ (12 electrons around S in the zero-formal-charge representation), and coordination compounds. (Note: Sulphur also obeys the octet rule in compounds like $\mathrm{SCl_2}$, $:\mathrm{Cl-\ddot{S}-Cl}:$, which has 8 electrons around S).
4. **Other Drawbacks of Octet Theory:**
   - It is based on the chemical inertness of noble gases, yet xenon and krypton combine with oxygen and fluorine to form compounds like $\mathrm{XeF_2, KrF_2, XeOF_2}$.
   - It does not account for the **shape** of molecules.
   - It is totally silent about the **energy** of a molecule and cannot explain relative stabilities.

3. Ionic or Electrovalent Bond & Lattice Enthalpy

The formation of ionic compounds primarily depends upon:

1. The ease of formation of positive and negative ions from their respective neutral gaseous atoms:

$$\mathrm{M(g)\rightarrow M^{+}(g)+e^-};\quad \text{Ionization enthalpy } (\Delta_i H,\ \text{always endothermic})$$

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$$\mathrm{X(g) + e^- \rightarrow X^-(g);} \quad \text{Electron gain enthalpy } (\Delta_{eg}H, \text{ may be exothermic or endothermic})$$
$$\mathrm{M^+(g) + X^-(g) \rightarrow MX(s)}$$

(Note: Electron affinity is the negative of the energy change accompanying electron gain.)

2. The three-dimensional arrangement of positive and negative ions in the solid crystal lattice.

Favourable Conditions for Ionic Bond Formation: Ionic bonds are formed more easily between elements with **comparatively low ionization enthalpies** and elements with **comparatively high negative values of electron gain enthalpy**. Most cations come from metallic elements and anions from non-metallic elements (an important exception is the ammonium ion, $\mathrm{NH_4^+}$, made of two non-metallic elements, which forms the cation of many ionic compounds).

3.1 Energetics of Crystal Stability & Lattice Enthalpy

In ionic solids, the sum of ionization enthalpy and electron gain enthalpy may be positive (endothermic), yet the crystal structure is stabilized by the large energy released during crystal lattice formation. For example, in $\mathrm{NaCl(s)}$:

- Ionization enthalpy for $\mathrm{Na(g) \rightarrow Na^+(g) + e^-}$ is $+495.8\,\mathrm{kJ\,mol^{-1}}$.
- Electron gain enthalpy for $\mathrm{Cl(g) + e^- \rightarrow Cl^-(g)}$ is $-348.7\,\mathrm{kJ\,mol^{-1}}$.
- The sum of the two is $+147.1\,\mathrm{kJ\,mol^{-1}}$, which is more than compensated for by the **enthalpy of lattice formation** of $\mathrm{NaCl(s)}$ $(-788\,\mathrm{kJ\,mol^{-1}})$.

Thus, a qualitative measure of the stability of an ionic compound is provided by its **enthalpy of lattice formation** and not simply by achieving an octet of electrons around the ionic species in the gaseous state.

Definition of Lattice Enthalpy: The Lattice Enthalpy of an ionic solid is defined as the energy required to completely separate one mole of a solid ionic compound into gaseous constituent ions to an infinite distance.

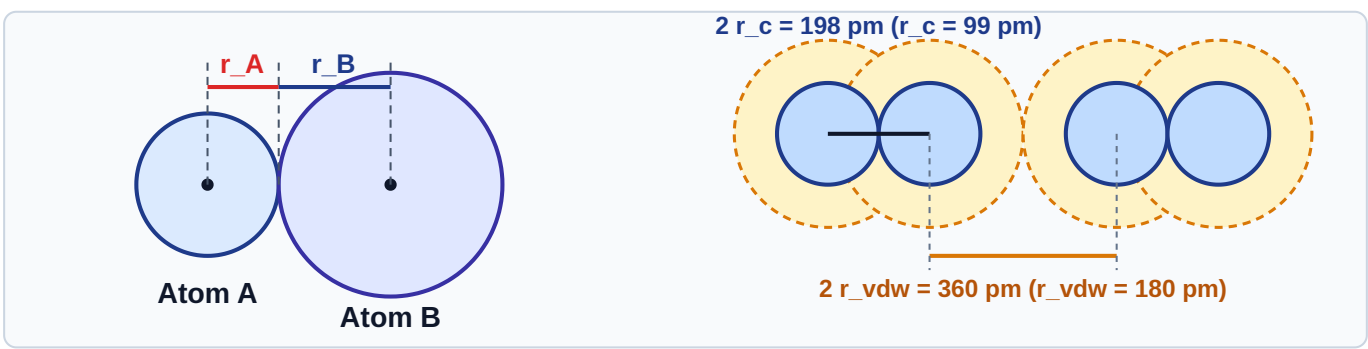
$$\mathrm{NaCl(s) \rightarrow Na^+(g) + Cl^-(g);} \quad \Delta_{\text{lattice}}H^{\ominus} = +788\,\mathrm{kJ\,mol^{-1}}$$

Because a solid crystal is three-dimensional, lattice enthalpy involves both attractive forces between oppositely charged ions and repulsive forces between like-charged ions, so crystal geometry factors must be included in its calculation.

4. Bond Parameters, Resonance, and Polarity of Bonds

4.1 Bond Length, Covalent Radius, and van der Waals Radius

- **Bond Length (R):** The equilibrium distance between the nuclei of two bonded atoms in a molecule, measured by spectroscopic, X-ray diffraction, and electron-diffraction techniques. For a covalent molecule AB, $R = r_\mathrm{A} + r_\mathrm{B}$, where r_A and r_B are the covalent radii of atoms A and B.
- **Covalent Radius (r_{cov} or r_c):** The radius of an atom's core which is in contact with the core of an adjacent atom in a bonded situation. It is half of the distance between two similar atoms joined by a covalent bond in the same molecule.
- **van der Waals Radius (r_{vdw}):** Represents the overall size of the atom including its valence shell in a non-bonded situation. It is half of the distance between two similar atoms in separate molecules in a solid. Always: $r_{\mathrm{vdw}} > r_{\mathrm{cov}}$ (for chlorine, $r_\mathrm{c} = 99\,\mathrm{pm}$ and $r_{\mathrm{vdw}} = 180\,\mathrm{pm}$).



Tutorial Diagram 2: (Left) Bond length $R = r_\mathrm{A} + r_\mathrm{B}$ in a covalent molecule AB. (Right) Covalent radius ($r_\mathrm{c} = 99\,\mathrm{pm}$) and van der Waals radius ($r_{\mathrm{vdw}} = 180\,\mathrm{pm}$) in chlorine ($\mathrm{Cl_2}$).

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Average Bond Lengths by Bond Type	Bond Lengths in Common Molecules	Covalent Radii r_{cov} (pm) (Bond order in parentheses)
$\text{O}-\text{H}: 96\,\text{pm}$ $\text{C}-\text{H}: 107\,\text{pm}$ $\text{N}-\text{O}: 136\,\text{pm}$ $\text{C}-\text{O}: 143\,\text{pm}$ $\text{C}-\text{N}: 143\,\text{pm}$ $\text{C}-\text{C}: 154\,\text{pm}$ $\text{C}=\text{O}: 121\,\text{pm}$ $\text{N}=\text{O}: 122\,\text{pm}$ $\text{C}=\text{C}: 133\,\text{pm}$ $\text{C}=\text{N}: 128\,\text{pm}$ $\text{C}\equiv\text{N}: 116\,\text{pm}$ $\text{C}\equiv\text{C}: 120\,\text{pm}$	$\text{H}_2\,(\text{H}-\text{H}): 74\,\text{pm}$ $\text{F}_2\,(\text{F}-\text{F}): 144\,\text{pm}$ $\text{Cl}_2\,(\text{Cl}-\text{Cl}): 199\,\text{pm}$ $\text{Br}_2\,(\text{Br}-\text{Br}): 228\,\text{pm}$ $\text{I}_2\,(\text{I}-\text{I}): 267\,\text{pm}$ $\text{N}_2\,(\text{N}\equiv\text{N}): 109\,\text{pm}$ $\text{O}_2\,(\text{O}=\text{O}): 121\,\text{pm}$ $\text{HF}\,(\text{H}-\text{F}): 92\,\text{pm}$ $\text{HCl}\,(\text{H}-\text{Cl}): 127\,\text{pm}$ $\text{HBr}\,(\text{H}-\text{Br}): 141\,\text{pm}$ $\text{HI}\,(\text{H}-\text{I}): 160\,\text{pm}$	$\text{H}: 37$ $\text{C}: 77(1),\ 67(2),\ 60(3)$ $\text{N}: 74(1),\ 65(2),\ 55(3)$ $\text{O}: 66(1),\ 57(2)$ $\text{F}: 64 \quad \text{Cl}: 99 \quad \text{Br}: 114 \quad \text{I}: 133$ $\text{P}: 110 \quad \text{As}: 121 \quad \text{Sb}: 141$ $\text{S}: 104(1),\ 95(2) \quad \text{Se}: 104 \quad \text{Te}: 137$

4.2 Bond Angle and Bond Enthalpy

Bond Angle: The angle between the orbitals containing bonding electron pairs around the central atom in a molecule/complex ion, expressed in degrees ($^\circ$) and determined experimentally by spectroscopic methods (e.g., the $\text{H}-\text{O}-\text{H}$ bond angle in water is 104.5°).

Bond Enthalpy ($\Delta_{\text{a}}H^\ominus$): The amount of energy required to break one mole of bonds of a particular type between two atoms in a gaseous state (unit: kJ mol^{-1}). Larger the bond dissociation enthalpy, stronger is the bond:

- $\text{H}_2(\text{g}) \rightarrow \text{H}(\text{g}) + \text{H}(\text{g}); \quad \Delta_{\text{a}}H^\ominus = 435.8\,\text{kJ mol}^{-1}$
- $\text{O}_2\,(\text{O}=\text{O})(\text{g}) \rightarrow \text{O}(\text{g}) + \text{O}(\text{g}); \quad \Delta_{\text{a}}H^\ominus = 498\,\text{kJ mol}^{-1}$
- $\text{N}_2\,(\text{N}\equiv\text{N})(\text{g}) \rightarrow \text{N}(\text{g}) + \text{N}(\text{g}); \quad \Delta_{\text{a}}H^\ominus = 946.0\,\text{kJ mol}^{-1}$
- $\text{HCl}(\text{g}) \rightarrow \text{H}(\text{g}) + \text{Cl}(\text{g}); \quad \Delta_{\text{a}}H^\ominus = 431.0\,\text{kJ mol}^{-1}$

Mean or Average Bond Enthalpy in Polyatomic Molecules: In polyatomic molecules like H_2O, the enthalpy needed to break the two $\text{O}-\text{H}$ bonds successively is not the same because the chemical environment changes after breaking the first bond:

$$\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}(\text{g}) + \text{OH}(\text{g}); \quad \Delta_{\text{a}}H_1^\ominus = 502\,\text{kJ mol}^{-1}$$
$$\text{OH}(\text{g}) \rightarrow \text{H}(\text{g}) + \text{O}(\text{g}); \quad \Delta_{\text{a}}H_2^\ominus = 427\,\text{kJ mol}^{-1}$$
$$\text{Average Bond Enthalpy} = \frac{\text{Total bond dissociation enthalpy}}{\text{Number of bonds broken}} = \frac{502 + 427}{2} = 464.5\,\text{kJ mol}^{-1}$$

4.3 Bond Order

In the Lewis description, **Bond Order** is given by the number of shared electron pairs (bonds) between two atoms in a molecule: H_2 ($\text{B.O.} = 1$), O_2 ($\text{B.O.} = 2$), N_2 ($\text{B.O.} = 3$), and CO ($\text{B.O.} = 3$).

Key Correlations for NEET UG:

- **Isoelectronic molecules and ions have identical bond orders:**
  - F_2 and O_2^{2-} (18 electrons each) have $\text{Bond Order} = 1$.
  - N_2, CO, and NO^+ (14 electrons each) have $\text{Bond Order} = 3$.
- **With increase in bond order $\implies$ bond enthalpy increases and bond length decreases.**

4.4 Resonance Structures

Whenever a single Lewis structure cannot describe a molecule accurately in conformity with its experimentally determined parameters, a number of structures with similar energy, positions of nuclei, and bonding and non-bonding pairs of electrons are taken as the **canonical structures (or resonance structures)** of the **resonance hybrid** which describes the molecule accurately. Resonance is represented by a double-headed arrow ($\leftrightarrow$).

- **Ozone (O_3):** Normal $\text{O}-\text{O}$ single bond length is $148\,\text{pm}$ and normal $\text{O}=\text{O}$ double bond length is $121\,\text{pm}$, but experimentally both $\text{O}-\text{O}$ bonds in O_3 have the same length ($128\,\text{pm}$), intermediate between a single and double bond.
- **Carbonate Ion (CO_3^{2-}):** All three carbon-to-oxygen bonds are experimentally equivalent, represented as a hybrid of three canonical forms.

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- **Carbon Dioxide (CO_2):** Normal $C=O$ is $121\,pm$ and normal $C\equiv O$ is $110\,pm$. Experimentally, the $C-O$ bond length in CO_2 is $115\,pm$, described as a resonance hybrid of $:\ddot{O}=C=\ddot{O}: \leftrightarrow :\ddot{O}:^- -C\equiv O:^+ \leftrightarrow :O^+\equiv C-\ddot{O}:^-$.

Essential Facts & Misconceptions about Resonance:

- Resonance **stabilizes** the molecule because the energy of the resonance hybrid is less than the energy of any single canonical structure, and resonance **averages the bond characteristics** as a whole.
- The canonical forms have **no real existence**.
- The molecule does *not* exist for a certain fraction of time in one canonical form and for other fractions of time in other canonical forms.
- There is *no equilibrium* between canonical forms (unlike keto-enol tautomerism).
- Canonicals must have the **same positions of nuclei (atomic connectivity)**; structures that differ in atomic connectivity (such as H_3PO_3 structures with $P-H$ vs. $O-H$ connectivity) are *not* resonance canonicals.

4.5 Polarity of Bonds, Dipole Moment (μ), and Fajans' Rules

No bond or compound is 100% covalent or 100% ionic. In homonuclear molecules (H_2, O_2, Cl_2, N_2, F_2), the shared electron pair is equally attracted by both identical nuclei, forming a **nonpolar covalent bond**. In heteronuclear molecules like HF, the shared pair is displaced toward the more electronegative fluorine atom, forming a **polar covalent bond**.

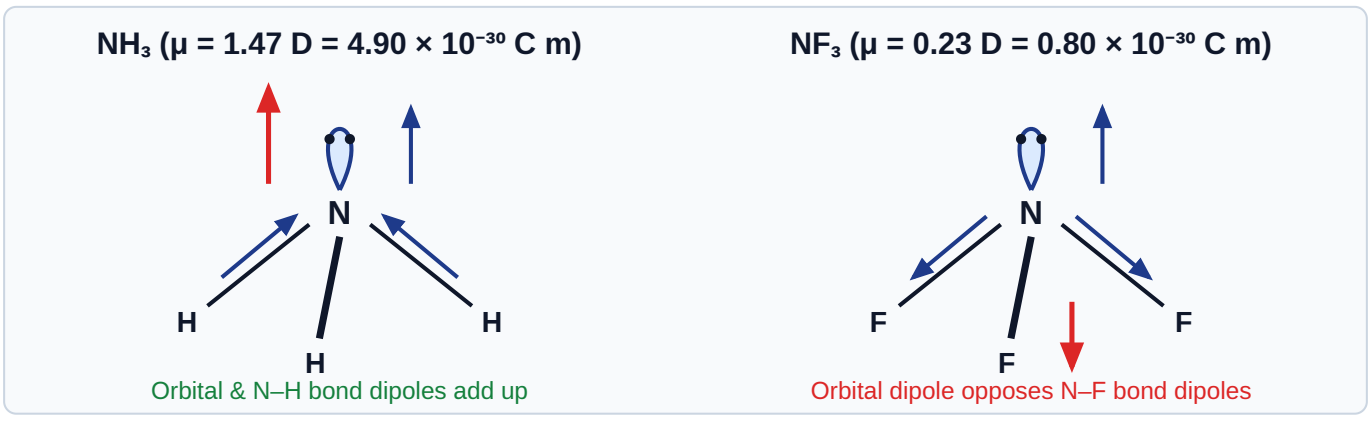
Dipole Moment (μ): The product of the magnitude of the charge (Q) and the distance (r) between the centres of positive and negative charge:

$$\mu = Q \times r$$

Unit: Debye (D), where $1\,D = 3.33564\times10^{-30}\,C\,m$. In chemistry, bond dipole is represented by a crossed arrow ($\longmapsto$) with the cross on the positive end and the arrowhead on the negative end, symbolizing the direction of electron density shift.

In polyatomic molecules, the net dipole moment is the **vector sum** of individual bond dipoles and lone-pair orbital dipoles:

- H_2O **(Bent, 104.5°):** The two $O-H$ bond dipoles reinforce each other to give a net $\mu = 1.85\,D = 6.17\times10^{-30}\,C\,m$.
- BeF_2 / BeH_2 / CO_2 **(Linear, 180°):** Two equal bond dipoles point in opposite directions and cancel out ($\mu = 0$).
- BF_3 **(Trigonal planar, 120°):** Three equal $B-F$ bond moments at 120° give a vector sum of zero ($\mu = 0$) because the resultant of any two is equal and opposite to the third.
- NH_3 **vs.** NF_3 **(Both trigonal pyramidal with 1 lone pair on N):** Although fluorine is more electronegative than nitrogen, $\mu(NH_3) = 1.47\,D$ ($4.90\times10^{-30}\,C\,m$) is much greater than $\mu(NF_3) = 0.23\,D$ ($0.80\times10^{-30}\,C\,m$). In NH_3, the orbital dipole due to the lone pair is in the **same direction** as the resultant dipole moment of the three $N-H$ bonds, whereas in NF_3, the lone-pair orbital dipole points in the **direction opposite** to the resultant dipole moment of the three $N-F$ bonds, partially cancelling it.



Tutorial Diagram 3: Comparison of bond dipoles and lone-pair orbital dipoles in NH_3 and NF_3.

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Molecule Type	Example	Dipole Moment μ (D)	Molecular Geometry
AB	HF, HCl, HBr, HI, H_2	1.78, 1.07, 0.79, 0.38, 0	Linear
AB_2	H_2O, H_2S, CO_2	1.85, 0.95, 0	Bent (H_2O, H_2S), Linear (CO_2)
AB_3	NH_3, NF_3, BF_3	1.47, 0.23, 0	Trigonal-pyramidal (NH_3, NF_3), Trigonal-planar (BF_3)
AB_4	CH_4, $CHCl_3$, CCl_4	0, 1.04, 0	Tetrahedral

Fajans' Rules (Partial Covalent Character of Ionic Bonds)

A cation polarises an anion by pulling electronic charge toward itself, increasing electron density between the nuclei. According to Fajans' rules, the **covalent character of an ionic bond increases** with:

1. **Smaller size of the cation** and **larger size of the anion** (greater polarising power of cation and greater polarisability of anion).
2. **Greater charge on the cation** (and anion).
3. **Pseudo-noble gas electronic configuration:** For cations of the same size and charge, a cation with electronic configuration $(n-1)d^n ns^0$ (typical of transition metals) is **more polarising** than one with a noble gas configuration ns^2np^6 (typical of alkali and alkaline earth metal cations).

5. The Valence Shell Electron Pair Repulsion (VSEPR) Theory

Proposed by Sidgwick and Powell (1940) and refined by Nyholm and Gillespie (1957), VSEPR theory predicts the shapes of covalent molecules based on repulsive interactions between electron pairs in the valence shell of the central atom.

- Electron pairs (bonded or non-bonded) around the central atom repel one another and occupy positions on the spherical valence shell surface that **minimise repulsion** and **maximise distance** between them.
- A multiple bond is treated as a **single electron pair (single super pair)**.
- Lone pairs are localised on the central atom alone, whereas bonding pairs are shared between two nuclei. Consequently, lone pairs occupy more space around the central atom, giving the fundamental order of repulsive interactions:

$$\text{Lone pair (lp)–Lone pair (lp)} > \text{Lone pair (lp)–Bond pair (bp)} > \text{Bond pair (bp)–Bond pair (bp)}$$

Molecule Type	Bonding Pairs	Lone Pairs	Arrangement of Electron Pairs	Molecular Geometry (Shape) & Bond Angles	Examples & Reason for Shape
AB_2	2	0	Linear	**Linear** (180°)	$BeCl_2$, $HgCl_2$, CO_2
AB_3	3	0	Trigonal planar	**Trigonal planar** (120°)	BF_3, BCl_3
AB_2E	2	1	Trigonal planar	**Bent (V-shaped)** (119.5° in SO_2)	SO_2, O_3 (lp–bp $>$ bp–bp reduces angle from 120°)
AB_4	4	0	Tetrahedral	**Tetrahedral** (109.5°)	CH_4, NH_4^+, $SiCl_4$
AB_3E	3	1	Tetrahedral	**Trigonal pyramidal** (107° in NH_3)	NH_3, PH_3 (lp–bp $>$ bp–bp reduces angle from 109.5°)
AB_2E_2	2	2	Tetrahedral	**Bent / Angular** (104.5° in H_2O)	H_2O, H_2S (lp–lp $>$ lp–bp $>$ bp–bp reduces angle to 104.5°)
AB_5	5	0	Trigonal bipyramidal	**Trigonal bipyramidal** (120° eq, 90° ax)	PCl_5, AsF_5
AB_4E	4	1	Trigonal bipyramidal	**See-saw** (distorted tetrahedron / folded square)	SF_4 (Lone pair occupies equatorial position with only two 90° lp–bp repulsions instead of three in axial)
AB_3E_2	3	2	Trigonal bipyramidal	**T-shaped**	ClF_3 (Both lone pairs occupy equatorial positions to minimise 90° repulsions)
AB_6	6	0	Octahedral	**Octahedral** (90°)	SF_6
AB_5E	5	1	Octahedral	**Square pyramidal**	BrF_5
AB_4E_2	4	2	Octahedral	**Square planar**	XeF_4 (Lone pairs are *trans* to each other)

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6. Valence Bond (VB) Theory & Hybridisation

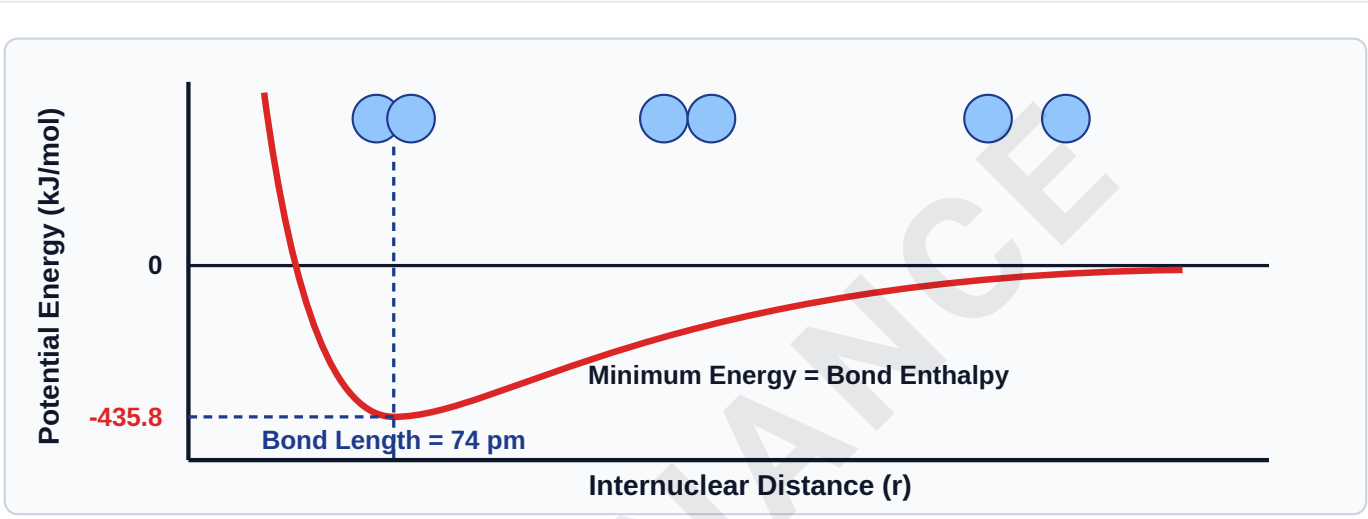
Introduced by Heitler and London (1927) and developed by Pauling, Valence Bond theory explains bond formation, bond dissociation enthalpies (e.g., H_2: $435.8\,kJ\,mol^{-1}$, $74\,pm$ vs. F_2: $155\,kJ\,mol^{-1}$, $144\,pm$), and directional properties using quantum mechanics.

6.1 Formation of H_2 Molecule and Potential Energy Curve

When two hydrogen atoms A and B (with nuclei N_A, N_B and electrons e_A, e_B) approach each other:

- **New attractive forces** arise between $N_A - e_B$ and $N_B - e_A$ (in addition to old $N_A - e_A$ and $N_B - e_B$).
- **New repulsive forces** arise between $e_A - e_B$ and $N_A - N_B$.

Because the magnitude of new attractive forces is greater than the new repulsive forces, the atoms approach each other and potential energy decreases until a minimum energy state is reached at an equilibrium internuclear distance (bond length) of $74\,pm$, releasing $435.8\,kJ\,mol^{-1}$ (bond enthalpy).



Tutorial Diagram 4: Potential energy curve for the formation of H_2 molecule as a function of internuclear distance.

6.2 Orbital Overlap Concept: Positive, Negative, and Zero Overlap

Greater the extent of partial interpenetration (overlapping) of half-filled atomic orbitals having opposite electron spins, stronger is the covalent bond formed. In boundary surface diagrams, **plus (+) and minus (−) signs indicate the sign (phase) of the orbital wave function (ψ) in space and are NOT related to electrical charge**:

- **Positive (in-phase) overlap:** Orbitals have the same sign (phase) and proper orientation (e.g., $p_z + p_z$ head-on, $s + p_z$ head-on, $p_x + p_x$ or $p_y + p_y$ sidewise with matching signs) $\implies$ bond formation occurs.
- **Negative (out-of-phase) overlap:** Orbitals approach with opposite signs (phases) $\implies$ no bond formation.
- **Zero overlap:** Out of phase due to different orientation/direction of approach (e.g., an s orbital approaching the nodal plane of p_x along the z-axis, or p_x and p_y orbitals approaching each other) $\implies$ net overlap is zero.

6.3 Sigma (σ) and Pi (π) Bonds

Feature	Sigma (σ) Bond	Pi (π) Bond
Type of Overlap	End-to-end (head-on / axial) overlap of bonding orbitals along the internuclear axis.	Sidewise (lateral) overlap of atomic orbitals whose axes are parallel to each other and perpendicular to the internuclear axis.
Orbital Combinations	s–s overlapping, s–p overlapping, or p–p axial overlapping (along internuclear axis).	p–p lateral overlapping (forms two saucer-type charged clouds above and below the molecular plane).
Extent of Overlap & Strength	Overlapping occurs to a larger extent; hence a σ **bond is stronger** than a π bond.	Overlapping occurs to a smaller extent; hence a π **bond is weaker** and is formed only in addition to a σ bond in multiple bonds.

6.4 Hybridisation of Atomic Orbitals

Pure atomic orbital overlap in CH_4, NH_3, and H_2O would predict 90° bond angles (from mutually perpendicular $2p_x, 2p_y, 2p_z$ orbitals), which disagrees with actual bond angles (109.5°, 107°, and 104.5°). Pauling introduced **hybridisation**: *the process of intermixing of the orbitals of slightly different energies so as to redistribute their energies, resulting in the formation of a new set of orbitals of equivalent energies and shape*.

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- **Salient Features:** (1) Number of hybrid orbitals equals the number of atomic orbitals hybridised; (2) Hybrid orbitals are always equivalent in energy and shape; (3) They form more stable bonds than pure orbitals; (4) They are directed in space to minimise repulsion, indicating molecular geometry.
- **Conditions:** (1) Valence-shell orbitals are hybridised; (2) Orbitals must have almost equal energy; (3) Promotion of electron is *not* an essential prior condition; (4) Both half-filled and completely filled valence orbitals (lone pairs) can participate.

Hybridisation	Orbitals Mixed & Character	Geometry & Angles	Key Examples & Structural Details
sp (Diagonal)	One s + one p ($50\%\ s, 50\%\ p$)	Linear (180°)	$BeCl_2$: Be ($1s^2 2s^2 \rightarrow 1s^2 2s^1 2p^1$) forms two sp hybrids overlapping axially with Cl $2p/3p$ orbitals. C_2H_2 **(Ethyne):** Both C atoms are sp hybridised; $C \equiv C$ has $1\,\sigma\,(sp\text{–}sp) + 2\,\pi\,(2p_x\text{–}2p_x, 2p_y\text{–}2p_y)$. Total bonds in C_2H_2: $3\,\sigma$ and $2\,\pi$.
sp^2 (Trigonal)	One s + two p ($33.3\%\ s, 66.7\%\ p$)	Trigonal planar (120°)	BCl_3: B ($1s^2 2s^2 2p^1 \rightarrow 1s^2 2s^1 2p_x^1 2p_y^1$) forms three sp^2 hybrids at 120°. C_2H_4 **(Ethene):** $C=C$ has $1\,\sigma\,(sp^2\text{–}sp^2) + 1\,\pi\,(2p\text{–}2p)$ (length 134 pm); four $C-H$ $\sigma\,(sp^2\text{–}s)$ bonds (108 pm); $\angle HCH = 117.6^\circ$, $\angle HCC = 121^\circ$. Total bonds: $5\,\sigma, 1\,\pi$.
sp^3 (Tetrahedral)	One s + three p ($25\%\ s, 75\%\ p$)	Tetrahedral (109.5°)	CH_4: Four $\sigma\,(sp^3\text{–}s)$ bonds at 109.5°. NH_3: Three $N-H$ bonds + 1 lone pair in sp^3 orbital ($\angle HNH = 107^\circ$, pyramidal). H_2O: Two $O-H$ bonds + 2 lone pairs in sp^3 orbitals ($\angle HOH = 104.5^\circ$, bent/V-shape). C_2H_6 **(Ethane):** One $C-C$ $\sigma\,(sp^3\text{–}sp^3)$ (154 pm) + six $C-H$ $\sigma\,(sp^3\text{–}s)$ (109 pm).
dsp^2	$d + s + p(2)$	Square planar (90°)	$[Ni(CN)_4]^{2-}$, $[PtCl_4]^{2-}$
sp^3d	$s + p(3) + d$	Trigonal bipyramidal ($120^\circ, 90^\circ$)	PF_5, PCl_5: Three **equatorial bonds** in one plane at 120°; two **axial bonds** above and below the plane at 90°. *Axial bonds suffer greater repulsion from three equatorial bond pairs at 90°, so axial bonds are slightly longer and weaker than equatorial bonds, making PCl_5 highly reactive.*
sp^3d^2 / d^2sp^3	$s + p(3) + d(2)$ or $d(2) + s + p(3)$	Octahedral (90°) (or Square pyramidal with 1 lp)	**Octahedral:** SF_6, $[CrF_6]^{3-}$ (sp^3d^2); $[Co(NH_3)_6]^{3+}$ (d^2sp^3). **Square pyramidal:** BrF_5 (sp^3d^2 with 5 bond pairs and 1 lone pair).

Note on d-orbital hybridisation in Period 3: The energy of $3d$ orbitals is comparable to $3s$ and $3p$ orbitals as well as to $4s$ and $4p$ orbitals, so hybridisation involving either $(3s, 3p, 3d)$ or $(3d, 4s, 4p)$ is possible. However, because the energy difference between $3p$ and $4s$ orbitals is significant, no hybridisation involving $3p, 3d$, and $4s$ orbitals is possible.

7. Molecular Orbital (MO) Theory

Developed by F. Hund and R.S. Mulliken (1932), MO theory describes electrons in molecules using polycentric molecular orbitals formed by the **Linear Combination of Atomic Orbitals (LCAO)** of comparable energies, proper symmetry about the molecular axis (z-axis by convention), and maximum overlap.

7.1 Bonding vs. Antibonding Molecular Orbitals

When two atomic orbitals ψ_A and ψ_B combine, two molecular orbitals are formed:

- **Bonding Molecular Orbital ($\sigma = \psi_A + \psi_B$ or π):** Formed by constructive interference (addition of wave functions). Electron density is concentrated *between* the nuclei, lowering nuclear repulsion, lowering energy below parent atomic orbitals, and stabilising the molecule.
- **Antibonding Molecular Orbital ($\sigma^* = \psi_A - \psi_B$ or π^*):** Formed by destructive interference (subtraction of wave functions). Contains a **nodal plane** (zero electron density) between the nuclei, raising energy above parent atomic orbitals and destabilising the molecule.
- **Symmetry around Bond Axis (z-axis):** σ MOs ($\sigma 1s, \sigma^* 1s, \sigma 2s, \sigma^* 2s, \sigma 2p_z, \sigma^* 2p_z$) are symmetrical around the bond axis, whereas π MOs ($\pi 2p_x, \pi^* 2p_x, \pi 2p_y, \pi^* 2p_y$) are not symmetrical around the bond axis because of positive lobes above and negative lobes below the molecular plane.

7.2 Energy Level Sequence of Molecular Orbitals

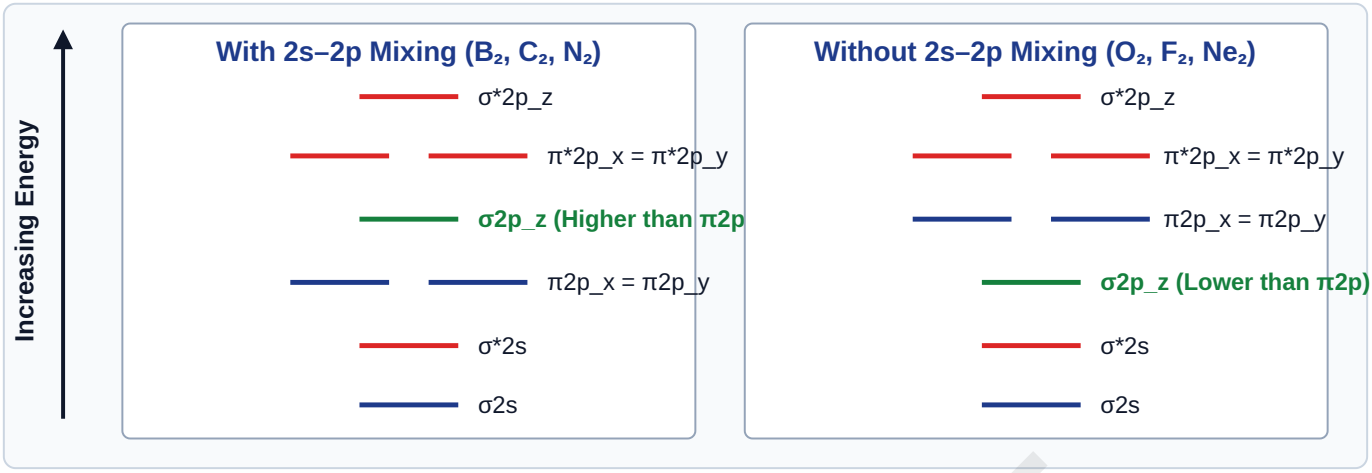
1. For B_2, C_2, N_2 (up to 14 electrons — with $2s$–$2p$ mixing):

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < (\pi 2p_x = \pi 2p_y) < \sigma 2p_z < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$$

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2. For O_2, F_2, Ne_2 (more than 14 electrons — without $2s$–$2p$ mixing):

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < (\pi 2p_x = \pi 2p_y) < (\pi^* 2p_x = \pi^* 2p_y) < \sigma^* 2p_z$$



Tutorial Diagram 5: Comparison of valence molecular orbital energy levels with $2s$–$2p$ mixing (B_2, C_2, N_2) and without $2s$–$2p$ mixing (O_2, F_2, Ne_2).

7.3 Bond Order, Stability, and Magnetic Behaviour

$$\text{Bond Order (b.o.)} = \frac{1}{2}(N_b - N_a)$$

where N_b = number of electrons in bonding orbitals and N_a = number of electrons in antibonding orbitals.

- **Stability:** Stable if $N_b > N_a$ (b.o. > 0); unstable and does not exist if $N_b \leq N_a$ (b.o. ≤ 0, e.g., He_2, Be_2, Ne_2).
- **Magnetic Nature:** If all MOs are doubly occupied, the species is **diamagnetic** (repelled by magnetic field). If one or more MOs are singly occupied (unpaired electrons), it is **paramagnetic** (attracted by magnetic field).

Molecule	Total e^-	Valence MO Configuration ($KK = (\sigma 1s)^2(\sigma^* 1s)^2$)	N_b, N_a	Bond Order	Bond Energy ($kJ\,mol^{-1}$) & Length (pm)	Magn Natur Speci
H_2	2	$(\sigma 1s)^2$	2, 0	1	$435.8\,kJ\,mol^{-1}$, 74 pm	Diama
He_2	4	$(\sigma 1s)^2(\sigma^* 1s)^2$	2, 2	0	Does not exist	Unsta
Li_2	6	$KK(\sigma 2s)^2$	4, 2	1	Exists in vapour phase	Diama
Be_2	8	$KK(\sigma 2s)^2(\sigma^* 2s)^2$	4, 4	0	Does not exist	Unsta
B_2	10	$KK(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^1 = \pi 2p_y^1)$	6, 4	1	$290\,kJ\,mol^{-1}$, 159 pm	**Paran** (2 unp e^- in
C_2	12	$KK(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^2 = \pi 2p_y^2)$	8, 4	2	$620\,kJ\,mol^{-1}$, 131 pm	**Diama** doubl consi **both**
N_2	14	$KK(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^2 = \pi 2p_y^2)(\sigma 2p_z)^2$	10, 4	3	$945\,kJ\,mol^{-1}$, 110 pm	Diama
O_2	16	$KK(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2(\pi 2p_x^2 = \pi 2p_y^2)(\pi^* 2p_x^1 = \pi^* 2p_y^1)$	10, 6	2	$498\,kJ\,mol^{-1}$, 121 pm	**Paran** (2 unp e^- in
F_2	18	$KK(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2(\pi 2p_x^2 = \pi 2p_y^2)(\pi^* 2p_x^2 = \pi^* 2p_y^2)$	10, 8	1	$159\,kJ\,mol^{-1}$, 143 pm	Diama
Ne_2	20	$KK(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2(\pi 2p_x^2 = \pi 2p_y^2)(\pi^* 2p_x^2 = \pi^* 2p_y^2)(\sigma^* 2p_z)^2$	10, 10	0	Does not exist	Unsta

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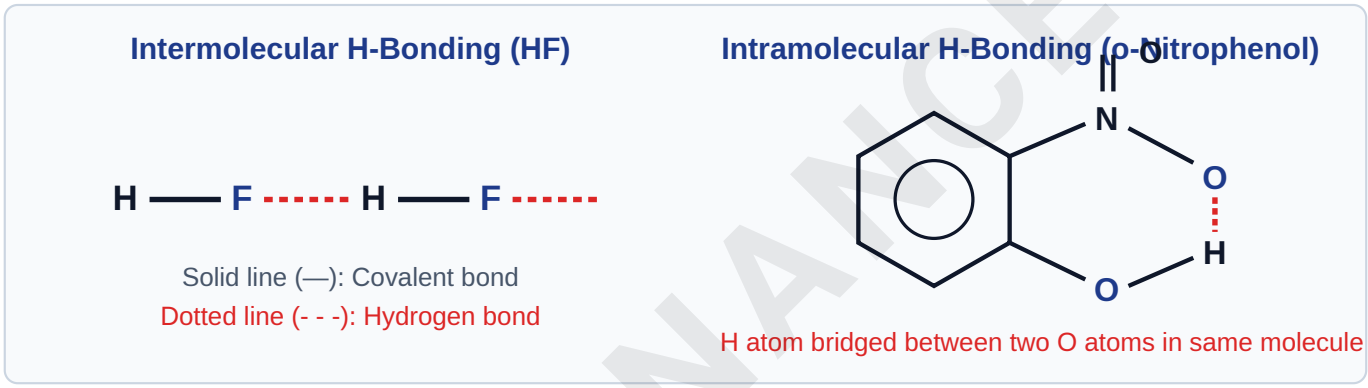
8. Hydrogen Bonding

Definition: The attractive force which binds the hydrogen atom of one molecule with the strongly electronegative atom (F, O, or N) of another molecule (or within the same molecule). It is weaker than a covalent bond ($\mathrm{H}-\mathrm{X}$) but stronger than van der Waals forces.

- **Cause of Formation:** When hydrogen is covalently bonded to a strongly electronegative element X ($\mathrm{F}, \mathrm{O}, \mathrm{N}$), the shared electron pair shifts toward X, giving hydrogen a fractional positive charge ($\mathrm{H}^{\delta+}$) and X a fractional negative charge ($\mathrm{X}^{\delta-}$):

$$\cdots \mathrm{H}^{\delta+}-\mathrm{F}^{\delta-} \cdots \mathrm{H}^{\delta+}-\mathrm{F}^{\delta-} \cdots \mathrm{H}^{\delta+}-\mathrm{F}^{\delta-}$$

- **Physical State Dependence:** The magnitude of H-bonding is **maximum in the solid state** and **minimum in the gaseous state**.
- **Types of H-Bonds:**
  1. **Intermolecular Hydrogen Bond:** Formed between two different molecules of the same or different compounds (e.g., HF, alcohol, or water molecules).
  2. **Intramolecular Hydrogen Bond:** Formed when a hydrogen atom is in between two highly electronegative ($\mathrm{F}, \mathrm{O}, \mathrm{N}$) atoms present within the *same molecule* (e.g., in *o*-nitrophenol, the phenolic $-\mathrm{OH}$ hydrogen is held between the two oxygen atoms of the $-\mathrm{OH}$ and $-\mathrm{NO}_2$ groups).



Tutorial Diagram 6: Intermolecular hydrogen bonding in HF and intramolecular hydrogen bonding in o-nitrophenol.

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9. Comprehensive Worked Examples

Worked Example 1: Writing Lewis Structures of CO and NO_2^-

Problem: Write the step-by-step Lewis dot structures of (a) carbon monoxide (CO) and (b) the nitrite ion (NO_2^-).

Solution:

(a) Carbon Monoxide (CO):

1. *Total valence electrons:* $C\,(2s^2 2p^2) = 4$ and $O\,(2s^2 2p^4) = 6 \implies 4 + 6 = 10$ valence electrons.
2. *Skeletal structure:* $C \quad O$.
3. *Electron distribution:* Drawing a single bond between C and O and completing the octet on O leaves two electrons as a lone pair on C ($:\mathrm{C} - \ddot{\mathrm{O}}:$), which gives C only 4 electrons. To complete the octet on both C and O, we form a triple bond between C and O: $:\mathrm{C} :::\mathrm{O}:$ or $:\mathrm{C} \equiv \mathrm{O}:$ (with formal charge -1 on C and $+1$ on O).

(b) Nitrite Ion (NO_2^-):

1. *Total valence electrons:* $N\,(2s^2 2p^3) = 5$, two $O\,(2s^2 2p^4) = 2 \times 6 = 12$, plus $1e^-$ for the -1 charge $\implies 5 + 12 + 1 = 18$ electrons.
2. *Skeletal structure:* $O \quad N \quad O$.
3. *Electron distribution:* Placing single bonds and completing octets on both oxygen atoms with a lone pair on nitrogen leaves nitrogen with only 6 electrons. Converting one oxygen lone pair into a $N = O$ double bond satisfies the octet rule for all three atoms: $[:\ddot{\mathrm{O}} = \ddot{\mathrm{N}} - \ddot{\mathrm{O}}:]^- \leftrightarrow [:\ddot{\mathrm{O}} - \ddot{\mathrm{N}} = \ddot{\mathrm{O}}:]^-$.

Worked Example 2: Correcting Skeletal Lewis Structures & Resonance Criteria

Problem: (a) An incorrect Lewis structure of acetic acid (CH_3COOH) is drawn with a $H = C$ double bond on the methyl carbon, a single $C - O$ bond to the carbonyl oxygen (with 4 non-bonding electrons on O), and a lone pair on the carboxyl carbon. Identify the errors and write the correct Lewis structure. (b) Phosphorous acid (H_3PO_3) is written in two forms: Form 1 has P bonded to one H, two $-OH$ groups, and one $:\mathrm{O}:$; Form 2 has P bonded to three $-OH$ groups and a lone pair on P. Can Form 1 and Form 2 be taken as canonical resonance forms of H_3PO_3 ? Explain.

Solution:

(a) Correct Lewis Structure of CH_3COOH: Hydrogen can only accommodate a duplet (1 covalent bond), so $H = C$ is impossible, and the methyl carbon would violate the octet rule with 5 bonds. In the correct Lewis structure, the methyl carbon forms three $C - H$ single bonds and one $C - C$ single bond; the carboxyl carbon forms a $C = O$ double bond (where O has 2 lone pairs) and a $C - O - H$ single bond (where the hydroxyl O has 2 lone pairs). Every C and O atom has a complete octet (8 electrons) and every H atom has a duplet (2 electrons).

(b) Canonical Forms of H_3PO_3: No, Form 1 and Form 2 cannot be taken as canonical resonance structures because canonical structures must have the **same positions of nuclei (identical atomic connectivity)** and differ only in the arrangement of electrons. In Form 1, one hydrogen atom is bonded directly to phosphorus ($P - H$), whereas in Form 2, all three hydrogen atoms are bonded to oxygen ($O - H$). Since the positions of the atoms have changed, they represent different structural isomers (tautomers), not resonance canonicals.

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Worked Example 3: Hybridisation Changes in Chemical Reactions & Organic Molecules

Problem: (a) Describe the change in hybridisation of the Al atom in $\mathrm{AlCl_3 + Cl^- \rightarrow AlCl_4^-}$ and of B and N atoms in $\mathrm{BF_3 + NH_3 \rightarrow F_3B \cdot NH_3}$. (b) State the hybridisation of each carbon atom from left to right in $\mathrm{CH_3 - CH = CH_2}$, $\mathrm{CH_3 - CHO}$, and $\mathrm{CH_3COOH}$.

Solution:

(a) Hybridisation in Adduct/Complex Formation:

- In $\mathrm{AlCl_3}$, Al has 3 bond pairs and 0 lone pairs, so Al is sp^2 hybridised (trigonal planar). In $\mathrm{AlCl_4^-}$, Al has 4 sigma bonds and 0 lone pairs, so the hybridisation of Al changes from sp^2 **to** sp^3 (tetrahedral).
- In $\mathrm{BF_3}$, B has 3 sigma bonds and 0 lone pairs (sp^2 hybridised). In $\mathrm{NH_3}$, N has 3 sigma bonds and 1 lone pair (sp^3 hybridised). Upon forming the adduct $\mathrm{F_3B \cdot NH_3}$, N donates its lone pair into the vacant $2p$ orbital of B to form a fourth sigma bond around B. Thus, the hybridisation of B changes from sp^2 **to** sp^3, while the hybridisation of N **remains** sp^3 (its lone pair becomes a bond pair, keeping 4 electron pairs around N).

(b) Carbon Hybridisations:

- $\mathrm{C_{(1)}H_3 - C_{(2)}H = C_{(3)}H_2}$: $\mathrm{C_1}$ has $4\,\sigma$ bonds (sp^3); $\mathrm{C_2}$ and $\mathrm{C_3}$ each have $3\,\sigma$ and $1\,\pi$ bond (sp^2).
- $\mathrm{C_{(1)}H_3 - C_{(2)}HO}$: $\mathrm{C_1}$ has $4\,\sigma$ bonds (sp^3); $\mathrm{C_2}$ (carbonyl carbon) has $3\,\sigma$ and $1\,\pi$ bond (sp^2).
- $\mathrm{C_{(1)}H_3 - C_{(2)}OOH}$: $\mathrm{C_1}$ has $4\,\sigma$ bonds (sp^3); $\mathrm{C_2}$ (carboxyl carbon) has $3\,\sigma$ and $1\,\pi$ bond (sp^2).

Worked Example 4: MO Theory Comparison of $\mathrm{O_2}$, $\mathrm{O_2^+}$, $\mathrm{O_2^-}$, and $\mathrm{O_2^{2-}}$

Problem: Using Molecular Orbital Theory, calculate the bond order, compare the relative stability and bond lengths, and indicate the magnetic properties of $\mathrm{O_2}$, $\mathrm{O_2^+}$, $\mathrm{O_2^-}$ (superoxide), and $\mathrm{O_2^{2-}}$ (peroxide).

Solution:

- $\mathrm{O_2^+}$ **(15 electrons):** Configuration has $N_\mathrm{b} = 10$ and $N_\mathrm{a} = 5$ (with 1 unpaired electron in π^*2p).

$$\text{Bond Order} = \frac{10 - 5}{2} = 2.5 \quad (\text{Paramagnetic})$$

- $\mathrm{O_2}$ **(16 electrons):** Configuration has $N_\mathrm{b} = 10$ and $N_\mathrm{a} = 6$ (with 2 unpaired electrons in $\pi^*2p_x^1, \pi^*2p_y^1$).

$$\text{Bond Order} = \frac{10 - 6}{2} = 2.0 \quad (\text{Paramagnetic})$$

- $\mathrm{O_2^-}$ **(Superoxide, 17 electrons):** Configuration has $N_\mathrm{b} = 10$ and $N_\mathrm{a} = 7$ (with 1 unpaired electron in $\pi^*2p_x^2, \pi^*2p_y^1$).

$$\text{Bond Order} = \frac{10 - 7}{2} = 1.5 \quad (\text{Paramagnetic})$$

- $\mathrm{O_2^{2-}}$ **(Peroxide, 18 electrons):** Configuration has $N_\mathrm{b} = 10$ and $N_\mathrm{a} = 8$ (all electrons paired: $\pi^*2p_x^2, \pi^*2p_y^2$).

$$\text{Bond Order} = \frac{10 - 8}{2} = 1.0 \quad (\text{Diamagnetic})$$

Order of Bond Order & Relative Stability: $\mathrm{O_2^+}\ (2.5) > \mathrm{O_2}\ (2.0) > \mathrm{O_2^-}\ (1.5) > \mathrm{O_2^{2-}}\ (1.0)$.

Order of Bond Length (inverse of bond order): $\mathrm{O_2^+ < O_2 < O_2^- < O_2^{2-}}$.

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10. NEET UG Practice Questions with Worked Solutions

Note: The following are original practice questions designed in strict alignment with the NEET UG examination pattern (Single-Correct MCQs, Statement-Based, Assertion-Reason, and Matrix-Match MCQs).

Practice Question 1 (Single-Correct MCQ — Formal Charge)

In the Lewis dot structure of the ozone molecule (O_3), represented as $\ddot{O}_{(2)} = \ddot{O}_{(1)} - \ddot{O}_{(3)}$:, the formal charges on the oxygen atoms numbered 1 (central), 2 (doubly bonded terminal), and 3 (singly bonded terminal) are respectively:

(1) $0,\ +1,\ -1$

(2) $+1,\ 0,\ -1$

(3) $-1,\ 0,\ +1$

(4) $+1,\ -1,\ 0$

Correct Answer: (2)

Worked Solution: Using $\text{F.C.} = V - L - \frac{1}{2}B$, where $V = 6$ for oxygen:

- For central $O_{(1)}$: $L = 2$ (one lone pair), $B = 6$ (three shared pairs) $\implies \text{F.C.} = 6 - 2 - \frac{1}{2}(6) = +1$.
- For doubly bonded $O_{(2)}$: $L = 4$ (two lone pairs), $B = 4$ (two shared pairs) $\implies \text{F.C.} = 6 - 4 - \frac{1}{2}(4) = 0$.
- For singly bonded $O_{(3)}$: $L = 6$ (three lone pairs), $B = 2$ (one shared pair) $\implies \text{F.C.} = 6 - 6 - \frac{1}{2}(2) = -1$.

Practice Question 2 (Single-Correct MCQ — Limitations of Octet Rule)

Which of the following sets contains species that represent, in order, an *incomplete octet*, an *odd-electron molecule*, and an *expanded octet* around the central atom?

(1) $BeH_2,\ NO_2,\ SF_6$

(2) $BF_3,\ CO,\ PF_5$

(3) $AlCl_3,\ NO,\ SCl_2$

(4) $LiCl,\ NF_3,\ H_2SO_4$

Correct Answer: (1)

Worked Solution: In BeH_2, the central Be atom is surrounded by only 4 valence electrons (incomplete octet). In NO_2, the total number of valence electrons is $5 + 2(6) = 17$, making it an odd-electron molecule. In SF_6, the central S atom is surrounded by 12 valence electrons (expanded octet). Note that in SCl_2, S has 2 bond pairs and 2 lone pairs (8 electrons, obeying the octet rule).

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Practice Question 3 (Single-Correct MCQ — Energetics of Ionic Bond Formation)

For the formation of $\mathrm{NaCl(s)}$, the ionization enthalpy of $\mathrm{Na(g)}$ is $+495.8\,\mathrm{kJ\,mol^{-1}}$, the electron gain enthalpy of $\mathrm{Cl(g)}$ is $-348.7\,\mathrm{kJ\,mol^{-1}}$, and the lattice enthalpy of $\mathrm{NaCl(s)}$ is $+788.0\,\mathrm{kJ\,mol^{-1}}$. What is the net enthalpy change when $1\,\mathrm{mol}$ of $\mathrm{Na(g)}$ and $1\,\mathrm{mol}$ of $\mathrm{Cl(g)}$ combine to form $1\,\mathrm{mol}$ of $\mathrm{NaCl(s)}$?

(1) $+147.1\,\mathrm{kJ\,mol^{-1}}$

(2) $-640.9\,\mathrm{kJ\,mol^{-1}}$

(3) $-935.1\,\mathrm{kJ\,mol^{-1}}$

(4) $-788.0\,\mathrm{kJ\,mol^{-1}}$

Correct Answer: (2)

Worked Solution: Adding the three steps:

1. $\mathrm{Na(g) \rightarrow Na^{+}(g) + e^{-}}$; $\quad \Delta_i H = +495.8\,\mathrm{kJ\,mol^{-1}}$
2. $\mathrm{Cl(g) + e^{-} \rightarrow Cl^{-}(g)}$; $\quad \Delta_{eg} H = -348.7\,\mathrm{kJ\,mol^{-1}}$
3. $\mathrm{Na^{+}(g) + Cl^{-}(g) \rightarrow NaCl(s)}$; $\quad \Delta_{\text{lattice form}} H = -788.0\,\mathrm{kJ\,mol^{-1}}$

$$\Delta H_{\text{net}} = (+495.8 - 348.7) - 788.0 = +147.1 - 788.0 = -640.9\,\mathrm{kJ\,mol^{-1}}$$

Practice Question 4 (Single-Correct MCQ — Isoelectronic Species & Bond Order)

Which of the following pairs of species is **not** isoelectronic and does **not** have the same bond order?

(1) $\mathrm{F_2}$ and $\mathrm{O_2^{2-}}$

(2) $\mathrm{N_2}$ and CO

(3) CO and $\mathrm{NO^{+}}$

(4) $\mathrm{O_2^{-}}$ and $\mathrm{NO^{+}}$

Correct Answer: (4)

Worked Solution:

- $\mathrm{F_2}$ ($18\,\mathrm{e^{-}}$) and $\mathrm{O_2^{2-}}$ ($18\,\mathrm{e^{-}}$) are isoelectronic with $\text{Bond Order} = 1$.
- $\mathrm{N_2}$ ($14\,\mathrm{e^{-}}$), CO ($14\,\mathrm{e^{-}}$), and $\mathrm{NO^{+}}$ ($7 + 8 - 1 = 14\,\mathrm{e^{-}}$) are isoelectronic with $\text{Bond Order} = 3$.
- $\mathrm{O_2^{-}}$ has $17\,\mathrm{e^{-}}$ ($\text{Bond Order} = 1.5$), whereas $\mathrm{NO^{+}}$ has $14\,\mathrm{e^{-}}$ ($\text{Bond Order} = 3.0$).

BOSONANCE

Practice Question 5 (Assertion-Reason — Dipole Moment of $\mathrm{NH_3}$ vs. $\mathrm{NF_3}$)

Assertion (A): Both $\mathrm{NH_3}$ and $\mathrm{NF_3}$ have pyramidal geometry with one lone pair on nitrogen, yet the resultant dipole moment of $\mathrm{NH_3}$ ($4.90 \times 10^{-30}\ \mathrm{C\,m}$) is much larger than that of $\mathrm{NF_3}$ ($0.80 \times 10^{-30}\ \mathrm{C\,m}$).
Reason (R): In $\mathrm{NH_3}$, the orbital dipole due to the lone pair is in the same direction as the resultant dipole moment of the three $\mathrm{N-H}$ bonds, whereas in $\mathrm{NF_3}$, the orbital dipole is in the direction opposite to the resultant dipole moment of the three $\mathrm{N-F}$ bonds.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

Correct Answer: (1)

Worked Solution: Nitrogen is more electronegative than hydrogen, so the three $\mathrm{N-H}$ bond dipoles point toward N, in the same direction as the lone-pair orbital dipole. Fluorine is more electronegative than nitrogen, so the three $\mathrm{N-F}$ bond dipoles point away from N, opposing the lone-pair orbital dipole and reducing the net dipole moment. Thus both (A) and (R) are true, and (R) correctly explains (A).

Practice Question 6 (Single-Correct MCQ — Fajans' Rules & Ionic Character)

Arrange the following molecules in order of **increasing ionic character** of their bonds: $\mathrm{LiF},\ \mathrm{K_2O},\ \mathrm{N_2},\ \mathrm{SO_2},\ \mathrm{ClF_3}$.

(1) $\mathrm{N_2} < \mathrm{SO_2} < \mathrm{ClF_3} < \mathrm{K_2O} < \mathrm{LiF}$

(2) $\mathrm{N_2} < \mathrm{ClF_3} < \mathrm{SO_2} < \mathrm{LiF} < \mathrm{K_2O}$

(3) $\mathrm{SO_2} < \mathrm{N_2} < \mathrm{ClF_3} < \mathrm{K_2O} < \mathrm{LiF}$

(4) $\mathrm{N_2} < \mathrm{SO_2} < \mathrm{ClF_3} < \mathrm{LiF} < \mathrm{K_2O}$

Correct Answer: (1)

Worked Solution: Ionic character increases with the electronegativity difference ($\Delta\mathrm{EN}$) between the bonded atoms and decreases with cation polarisation (Fajans' rules):

- $\mathrm{N_2}$ is homonuclear ($\Delta\mathrm{EN} = 0$), so it is nonpolar covalent (least ionic character).
- Among covalent polar molecules $\mathrm{SO_2}$ and $\mathrm{ClF_3}$, the $\mathrm{Cl-F}$ bond has a slightly larger electronegativity difference than $\mathrm{S=O}$ (due to the highest electronegativity of F), giving $\mathrm{SO_2} < \mathrm{ClF_3}$.
- $\mathrm{K_2O}$ and LiF are ionic solids. In LiF, the very high electronegativity of F (4.0) relative to Li (1.0) and the singly charged ions ($\mathrm{Li^+}, \mathrm{F^-}$) compared to the doubly charged oxide ion ($\mathrm{O^{2-}}$, which is more easily polarised) gives $\mathrm{N_2} < \mathrm{SO_2} < \mathrm{ClF_3} < \mathrm{K_2O} < \mathrm{LiF}$.

BOSONANCE

Practice Question 7 (Matrix-Match MCQ — VSEPR Molecular Shapes)

Match the molecules in **List I** with their VSEPR classification and molecular shape in **List II**:

List I (Molecule)	List II (VSEPR Type & Shape)
(A) $\mathrm{SF_4}$	(I) $\mathrm{AB_3E_2}$ – T-shaped
(B) $\mathrm{ClF_3}$	(II) $\mathrm{AB_4E_2}$ – Square planar
(C) $\mathrm{BrF_5}$	(III) $\mathrm{AB_4E}$ – See-saw
(D) $\mathrm{XeF_4}$	(IV) $\mathrm{AB_5E}$ – Square pyramidal

(1) (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

(2) (A)-(I), (B)-(III), (C)-(IV), (D)-(II)

(3) (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

(4) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

Correct Answer: (1)

Worked Solution:

- $\mathrm{SF_4}$: S has 6 valence $\mathrm{e^-}$ $\Longrightarrow$ 4 bond pairs + 1 lone pair ($\mathrm{AB_4E}$) $\Longrightarrow$ **See-saw** (III).
- $\mathrm{ClF_3}$: Cl has 7 valence $\mathrm{e^-}$ $\Longrightarrow$ 3 bond pairs + 2 lone pairs ($\mathrm{AB_3E_2}$) $\Longrightarrow$ **T-shaped** (I).
- $\mathrm{BrF_5}$: Br has 7 valence $\mathrm{e^-}$ $\Longrightarrow$ 5 bond pairs + 1 lone pair ($\mathrm{AB_5E}$) $\Longrightarrow$ **Square pyramidal** (IV).
- $\mathrm{XeF_4}$: Xe has 8 valence $\mathrm{e^-}$ $\Longrightarrow$ 4 bond pairs + 2 lone pairs ($\mathrm{AB_4E_2}$) $\Longrightarrow$ **Square planar** (II).

Practice Question 8 (Single-Correct MCQ — Orbital Overlap & Internuclear Axis)

Considering the x**-axis as the internuclear axis**, which of the following combinations of atomic orbitals will **NOT** form a sigma (σ) bond?

(1) $1s$ and $1s$

(2) $1s$ and $2p_x$

(3) $2p_y$ and $2p_y$

(4) $1s$ and $2s$

Correct Answer: (3)

Worked Solution: When the x-axis is taken as the internuclear axis, s orbitals (spherically symmetrical) and $2p_x$ orbitals (directed along the x-axis) undergo head-on (axial) overlap to form σ bonds. However, $2p_y$ orbitals have their axes perpendicular to the x-axis and parallel to each other; therefore, $2p_y$ and $2p_y$ undergo **sidewise (lateral) overlap** to form a **pi (π) bond**, not a sigma bond.

BOSONANCE

Practice Question 9 (Statement-Based MCQ — Resonance & Hybridisation)

Given below are two statements:
Statement I: $\mathrm{CH_4}$ is tetrahedral with a bond angle of 109.5° rather than square planar because in a square planar arrangement the bond angle is 90°, which causes much greater repulsion between the four bond pairs.
Statement II: In resonance, a molecule exists for a certain fraction of time in one canonical form and for other fractions of time in other canonical forms in rapid equilibrium.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct, but Statement II is incorrect.

(4) Statement I is incorrect, but Statement II is correct.

Correct Answer: (3)

Worked Solution: According to VSEPR theory, four electron pairs maximise distance and minimise repulsion at 109.5° (tetrahedral) compared to 90° (square planar), so Statement I is correct. Statement II is a classic misconception: canonical forms have no real existence, there is no equilibrium between them, and the molecule has a single resonance hybrid structure at all times.

Practice Question 10 (Assertion-Reason — Geometry & Reactivity of $\mathrm{PCl_5}$)

Assertion (A): In the trigonal bipyramidal $\mathrm{PCl_5}$ molecule, the two axial $\mathrm{P-Cl}$ bonds are slightly longer and weaker than the three equatorial $\mathrm{P-Cl}$ bonds, making $\mathrm{PCl_5}$ highly reactive.
Reason (R): The axial bond pairs suffer greater repulsive interaction from three equatorial bond pairs at 90° compared to the equatorial bond pairs, which have only two axial bond pairs at 90°.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) (A) is false, but (R) is true.

Correct Answer: (1)

Worked Solution: In $\mathrm{PCl_5}$ (sp^3d hybridisation), three equatorial $\mathrm{P-Cl}$ bonds lie in a plane at 120° to each other, while two axial $\mathrm{P-Cl}$ bonds lie perpendicular to the equatorial plane at 90°. Because 90° repulsions are much stronger than 120° repulsions, and each axial bond experiences three 90° repulsions (whereas each equatorial bond experiences only two 90° repulsions), the axial bonds elongate and weaken. Both (A) and (R) are true and (R) is the exact explanation of (A).

BOSONANCE

Practice Question 11 (Statement-Based MCQ — Molecular Orbital Theory of B_2 and C_2)

Given below are two statements regarding homonuclear diatomic molecules of the second period:
Statement I: In the vapour phase, the B_2 molecule is paramagnetic with bond order 1, whereas the C_2 molecule is diamagnetic with bond order 2.
Statement II: The double bond in the C_2 molecule consists of one sigma (σ) bond and one pi (π) bond.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct, but Statement II is incorrect.

(4) Statement I is incorrect, but Statement II is correct.

Correct Answer: (3)

Worked Solution: Due to $2s$–$2p$ mixing, $\pi 2p_x$ and $\pi 2p_y$ are lower in energy than $\sigma 2p_z$ for B_2 and C_2:

- B_2 ($10\,e^-$): $\mathrm{KK}(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^1 = \pi 2p_y^1) \implies \text{B.O.} = \frac{6-4}{2} = 1$, paramagnetic (2 unpaired e^-).
- C_2 ($12\,e^-$): $\mathrm{KK}(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x^2 = \pi 2p_y^2) \implies \text{B.O.} = \frac{8-4}{2} = 2$, diamagnetic. Thus Statement I is correct.
- Because the four net bonding electrons in C_2 occupy the two π molecular orbitals ($\pi 2p_x^2$ and $\pi 2p_y^2$) while $\sigma 2s^2$ and $\sigma^* 2s^2$ cancel out, **the double bond in C_2 consists of both pi (π) bonds**! Hence Statement II is incorrect.

Practice Question 12 (Single-Correct MCQ — Hydrogen Bonding & Bond Angles)

Which of the following statements is **correct** regarding hydrogen bonding and VSEPR bond angles?

(1) Although both NH_3 and H_2O have distorted tetrahedral (sp^3) geometries, the bond angle in H_2O (104.5°) is smaller than in NH_3 (107°) because H_2O has two lone pairs (lp–lp repulsion) whereas NH_3 has one lone pair.

(2) The magnitude of hydrogen bonding in a compound is maximum in the gaseous state and minimum in the solid state.

(3) *o*-Nitrophenol exhibits strong intermolecular hydrogen bonding, whereas HF exhibits intramolecular hydrogen bonding.

(4) A hydrogen bond is stronger than a normal covalent bond but weaker than van der Waals forces.

Correct Answer: (1)

Worked Solution:

- Option (1) is correct: H_2O (AB_2E_2) has two lone pairs, introducing strong lp–lp and lp–bp repulsions that compress the $H-O-H$ angle from 109.5° to 104.5°, whereas NH_3 (AB_3E) has only one lone pair (lp–bp repulsion), reducing the $H-N-H$ angle to 107°.
- Option (2) is false: H-bonding is maximum in the solid state and minimum in the gaseous state.
- Option (3) is false: *o*-Nitrophenol exhibits **intramolecular** H-bonding, while HF exhibits **intermolecular** H-bonding.
- Option (4) is false: A hydrogen bond is weaker than a covalent bond and stronger than van der Waals forces.

BOSONANCE

11. Common Pitfalls & Rapid Revision Checklist

High-Frequency Exam Pitfalls to Avoid

- **Confusing Orbital Sign ($+/-$) with Charge:** In orbital overlap diagrams, $+$ and $-$ signs represent the *phase (sign) of the wave function* ψ, never electrical charges.
- **Internuclear Axis Traps:** By default, the z-axis is the internuclear axis (so $2p_z + 2p_z \rightarrow \sigma$). However, if an exam question specifies the **x-axis** as the internuclear axis, then $2p_x + 2p_x \rightarrow \sigma$ while $2p_y + 2p_y \rightarrow \pi$ and $2p_z + 2p_z \rightarrow \pi$.
- **Position of Lone Pairs in $\mathrm{sp^3d}$ (TBP) vs. $\mathrm{sp^3d^2}$ (Octahedral):** In trigonal bipyramidal geometry ($\mathrm{SF_4}, \mathrm{ClF_3}, \mathrm{XeF_2}$), lone pairs *always occupy equatorial positions* to minimise 90° repulsions. In octahedral geometry ($\mathrm{XeF_4}$), two lone pairs occupy *trans* positions to stay 180° apart.
- **MO Energy Order Crossover at $14\,\mathrm{e^-}$:** Up to 14 electrons ($\mathrm{B_2}, \mathrm{C_2}, \mathrm{N_2}$), $\pi 2p_x = \pi 2p_y < \sigma 2p_z$. Above 14 electrons ($\mathrm{O_2}, \mathrm{F_2}, \mathrm{Ne_2}$), $\sigma 2p_z < \pi 2p_x = \pi 2p_y$. Never use the $\mathrm{O_2}$ order for $\mathrm{B_2}$ or $\mathrm{C_2}$ (which would wrongly predict $\mathrm{B_2}$ to be diamagnetic).
- **Nature of Double Bond in $\mathrm{C_2}$:** Unlike ethene ($1\,\sigma + 1\,\pi$), the double bond in $\mathrm{C_2}$ consists of **two π bonds and zero net σ bonds**.

Concise Final Recap

- **Formal Charge:** $\mathrm{F.C.} = V - L - \frac{1}{2}B$. Smallest formal charges correspond to the lowest-energy Lewis structure.
- **Dipole Moment:** $\mu = Q \times r$ ($1\,\mathrm{D} = 3.33564 \times 10^{-30}\,\mathrm{C\,m}$). Symmetrical molecules with no lone pairs ($\mathrm{BeF_2}, \mathrm{BeH_2}, \mathrm{CO_2}, \mathrm{BF_3}, \mathrm{CH_4}, \mathrm{CCl_4}$) and square planar $\mathrm{XeF_4}$ have $\mu = 0$. Crucially, $\mu(\mathrm{NH_3}) > \mu(\mathrm{NF_3})$.
- **Fajans' Rules:** Small cation + large anion + high ionic charge + $(n-1)d^n ns^0$ cation configuration $\implies$ maximum polarisation $\implies$ highest covalent character.
- **VSEPR Repulsion Order:** $\mathrm{lp–lp} > \mathrm{lp–bp} > \mathrm{bp–bp}$. Bond angles: $\mathrm{CH_4}\,(109.5^\circ) > \mathrm{NH_3}\,(107^\circ) > \mathrm{H_2O}\,(104.5^\circ)$.
- **MO Bond Order:** $\mathrm{b.o.} = \frac{1}{2}(N_\mathrm{b} - N_\mathrm{a})$. Higher bond order $\implies$ higher bond dissociation enthalpy and shorter bond length.

BOSONANCE